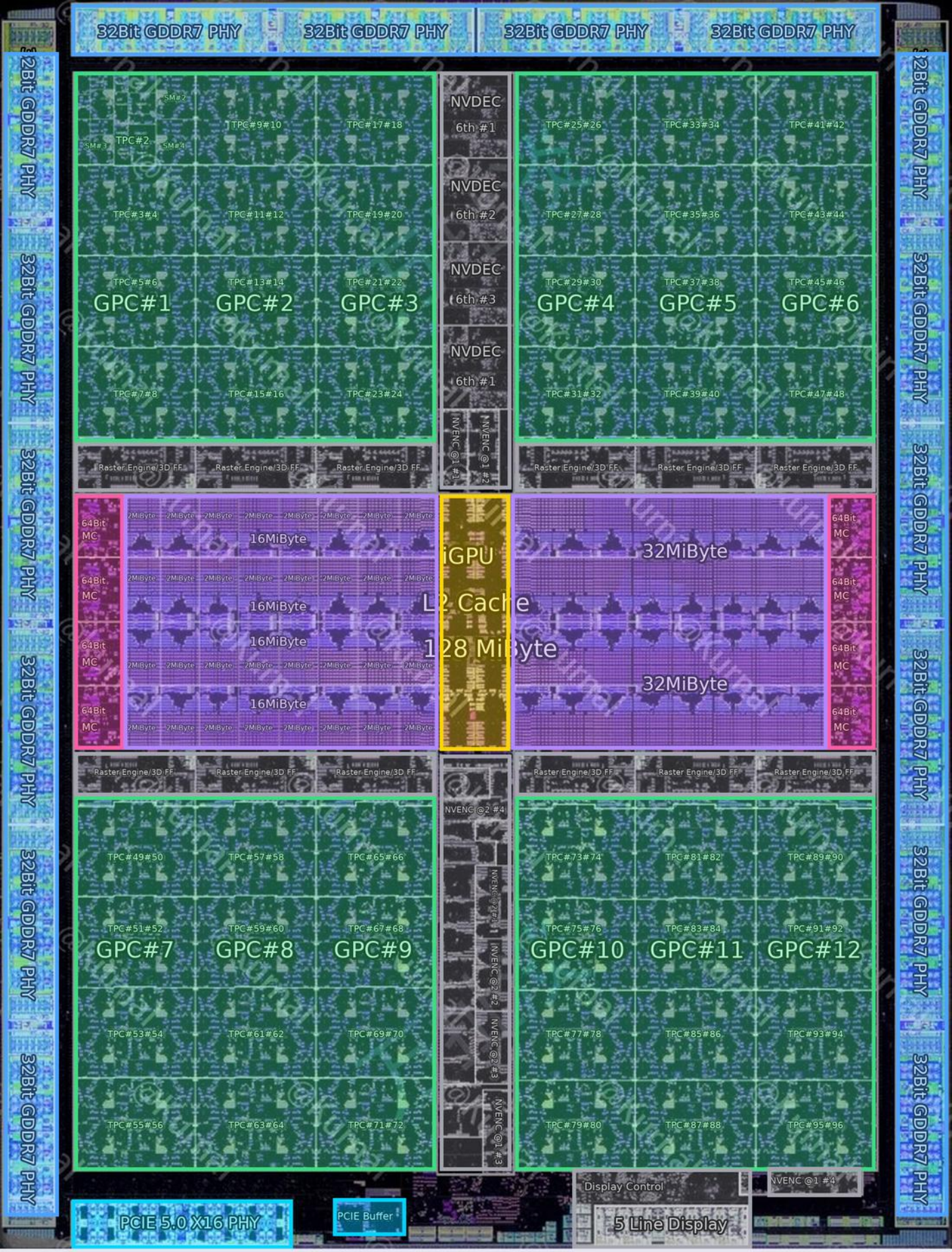
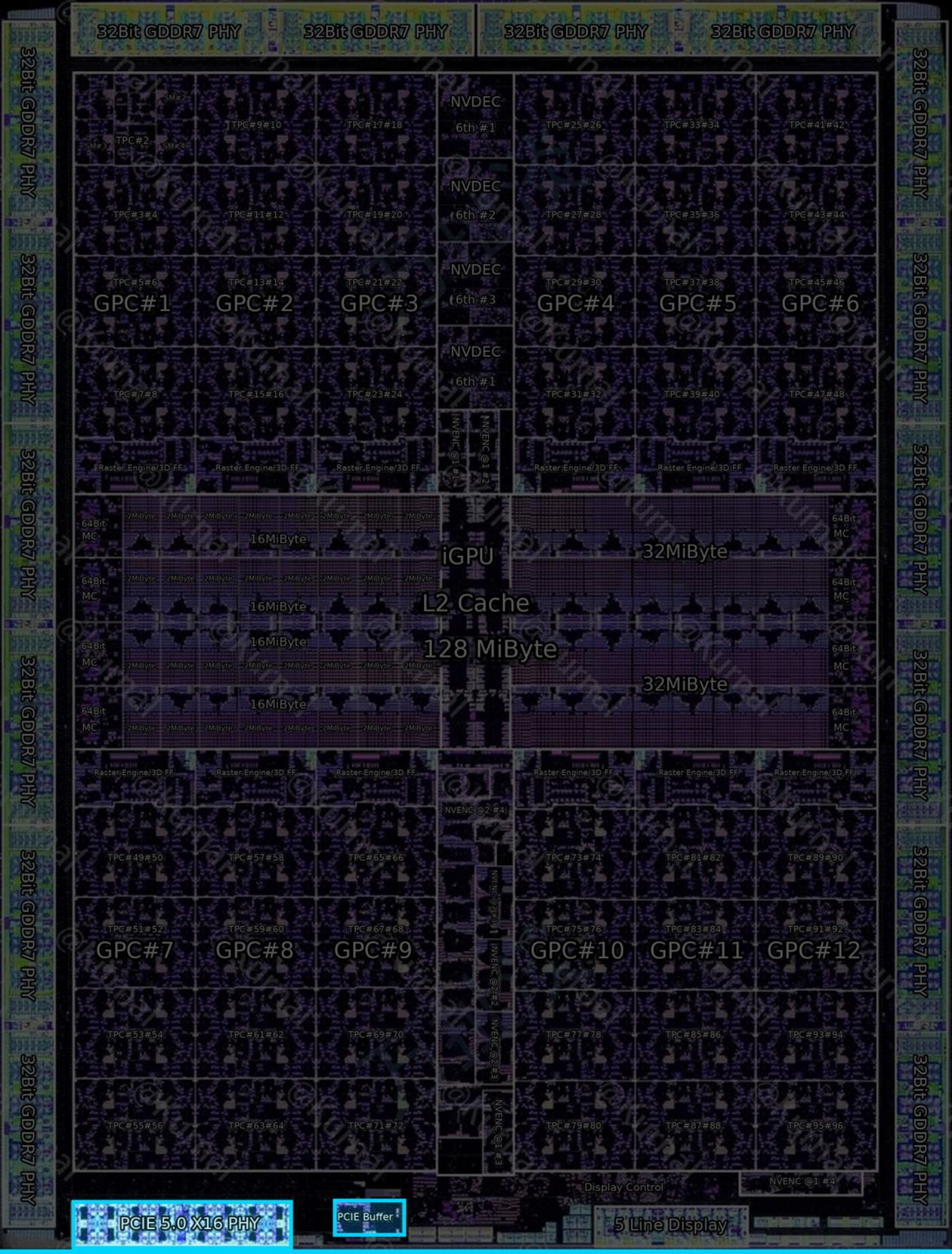


Overview

ML on GB202: who does the work

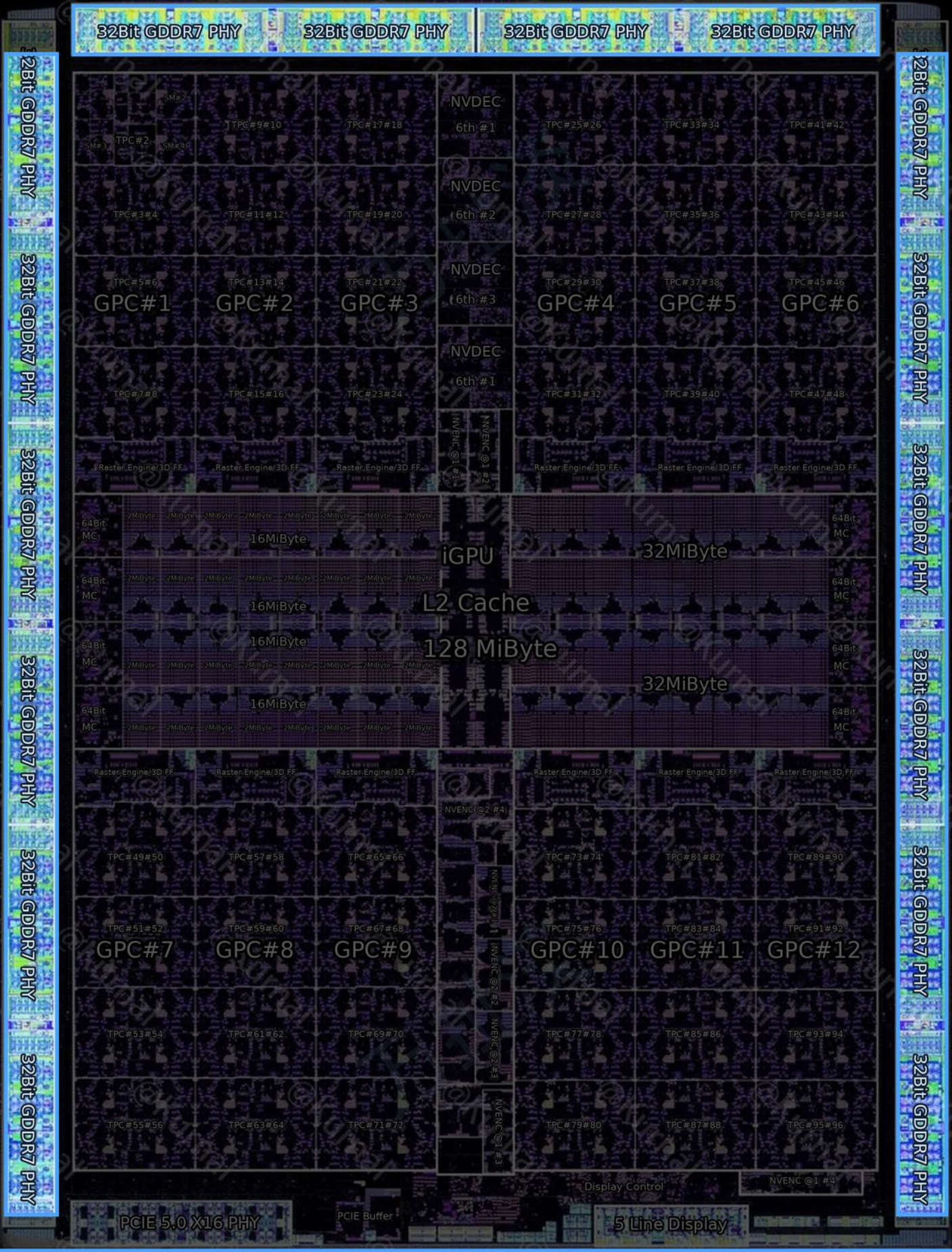
Colored blocks are busy during ML; gray blocks (raster engines, display, video) sit idle. This follows one simple case: a language model generating text. Roles marked (guess) are unconfirmed.



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Load the model

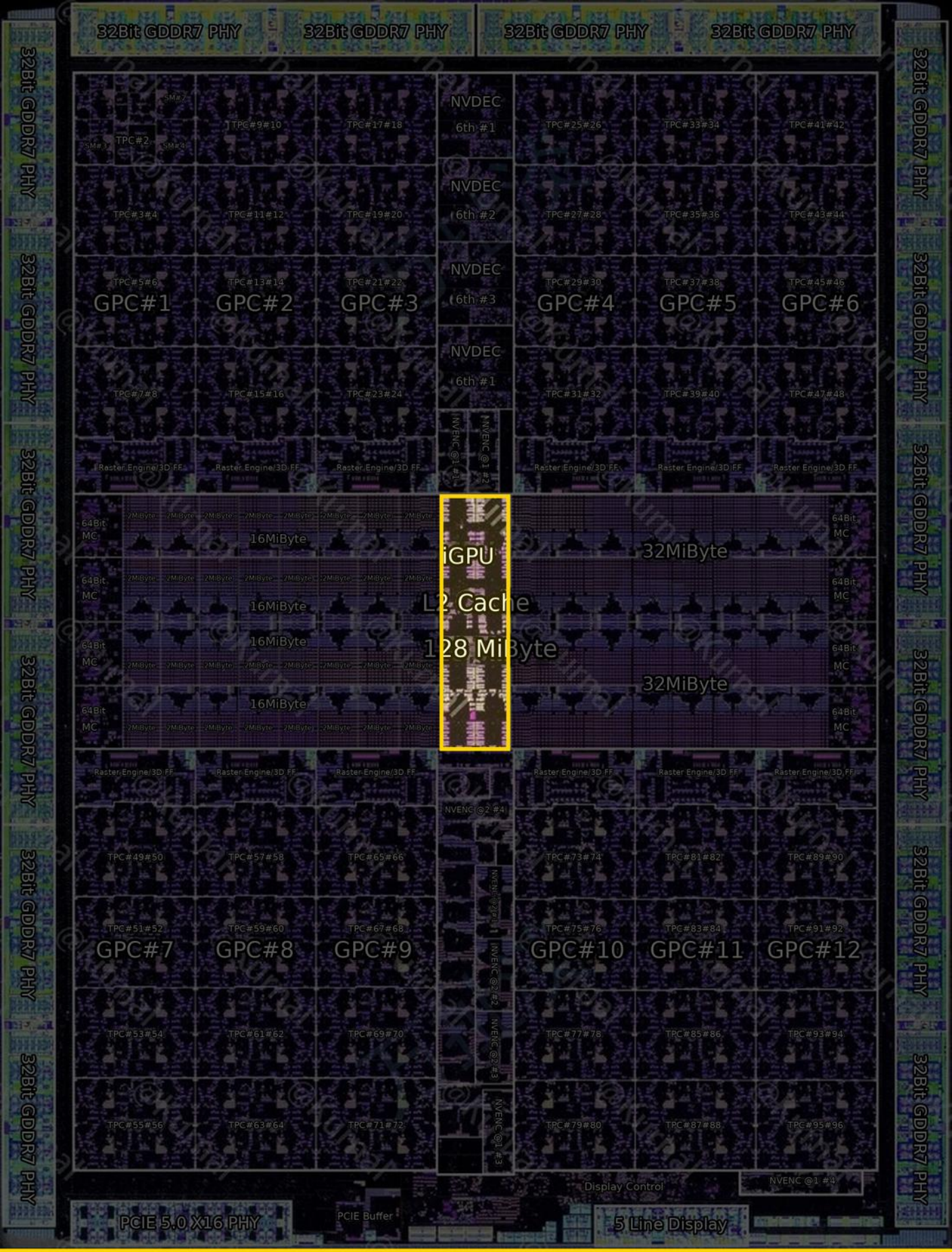
The model's weights (billions of numbers) come in from the computer over PCIe. This is a one-time trip. Once the model is loaded, PCIe is mostly quiet.



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The model lives in GDDR7

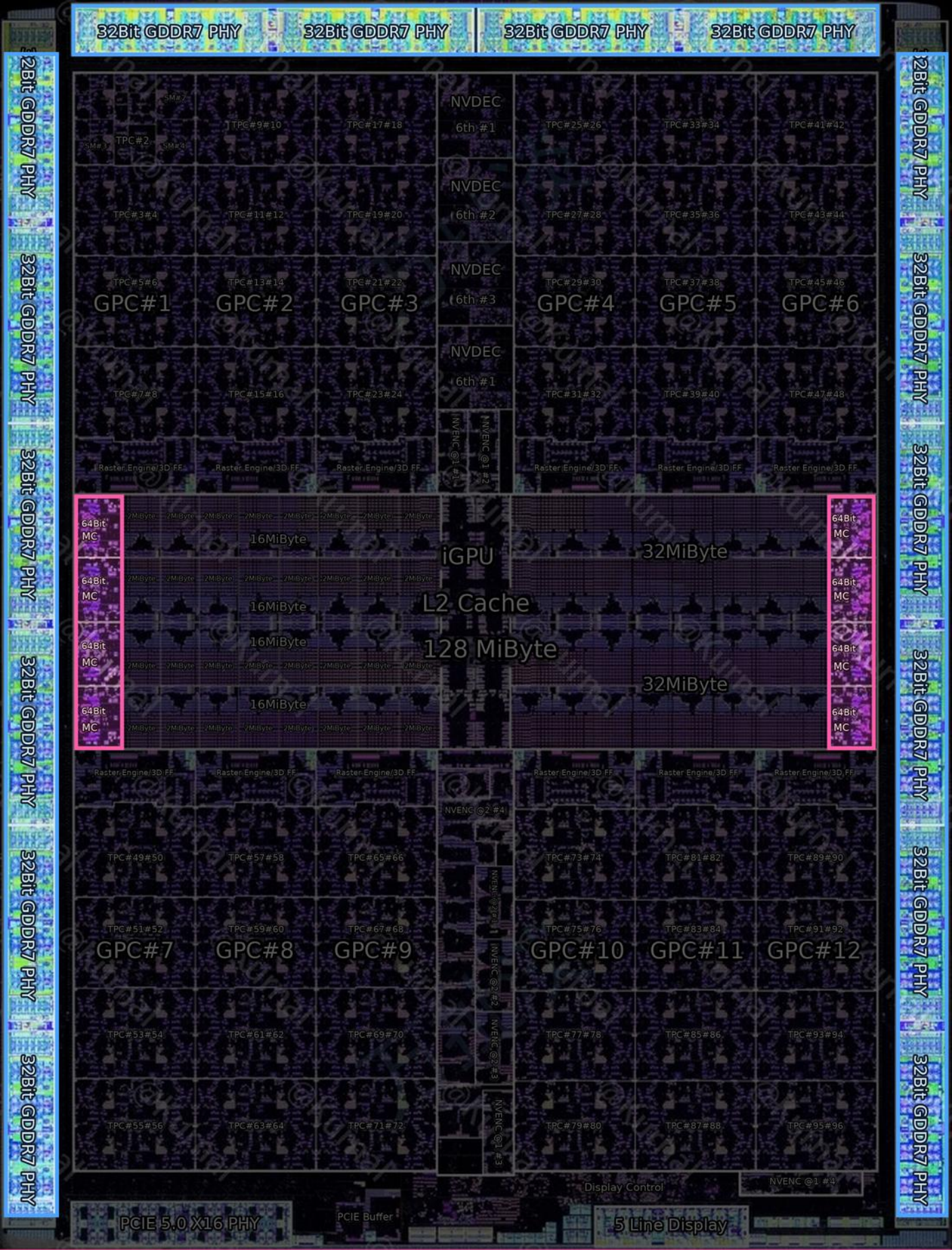
The weights are stored in the GDDR7 chips on the board (32 GB on the RTX 5090). These PHYs are the connection between those chips and the GPU.



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Jobs get handed out (guess)

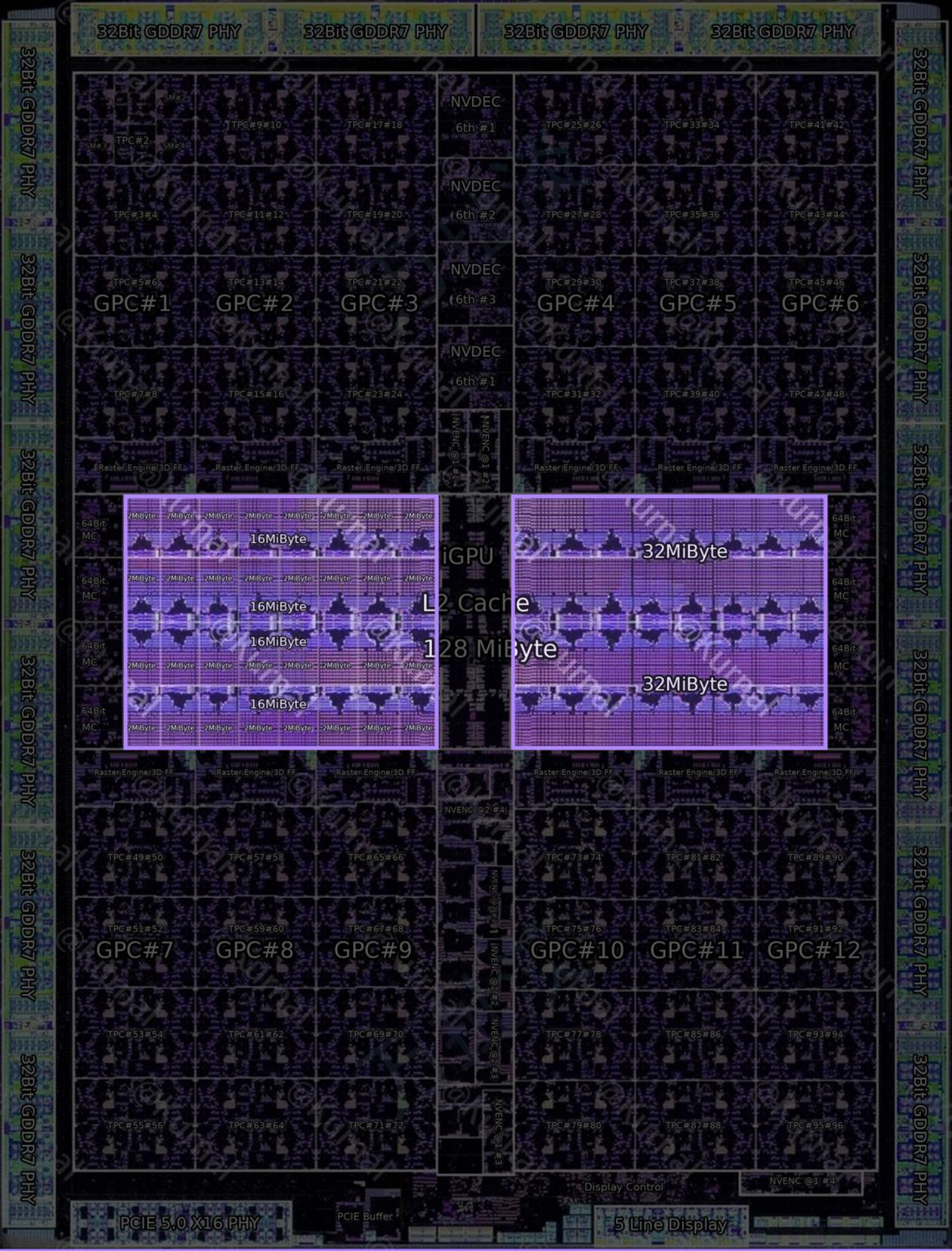
The GPU gets work as compute jobs, split into many small chunks and handed to SMs across all the GPCs. Nvidia describes a front-end processor that schedules this, but where it sits on the die is my guess.



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Weights stream in

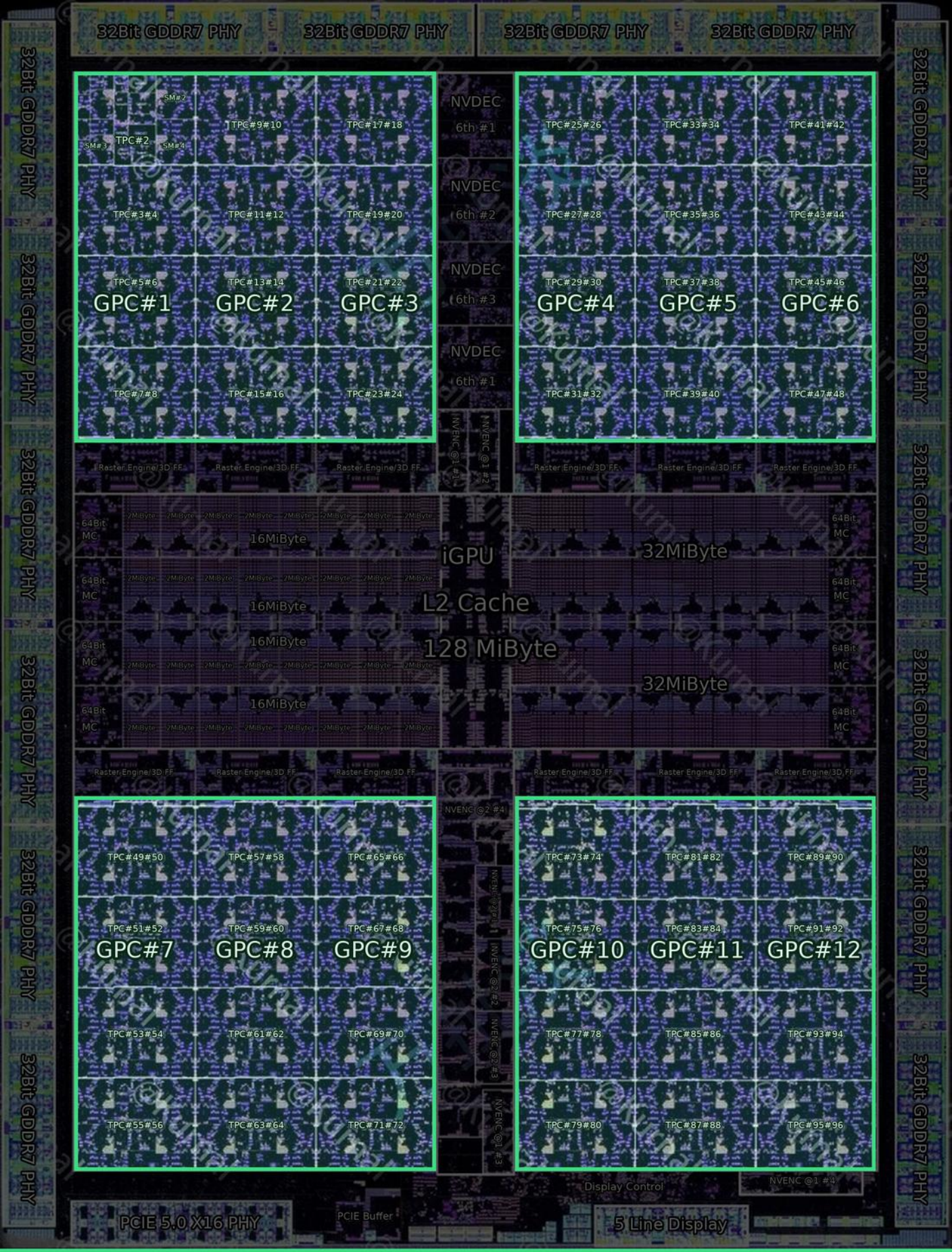
To produce each token (roughly a word), the GPU reads the model's weights from memory. At best this card moves about 1.8 TB/s (512 bits x 28 Gbps). Memory is usually the limit.



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L2 cache: the shared staging area

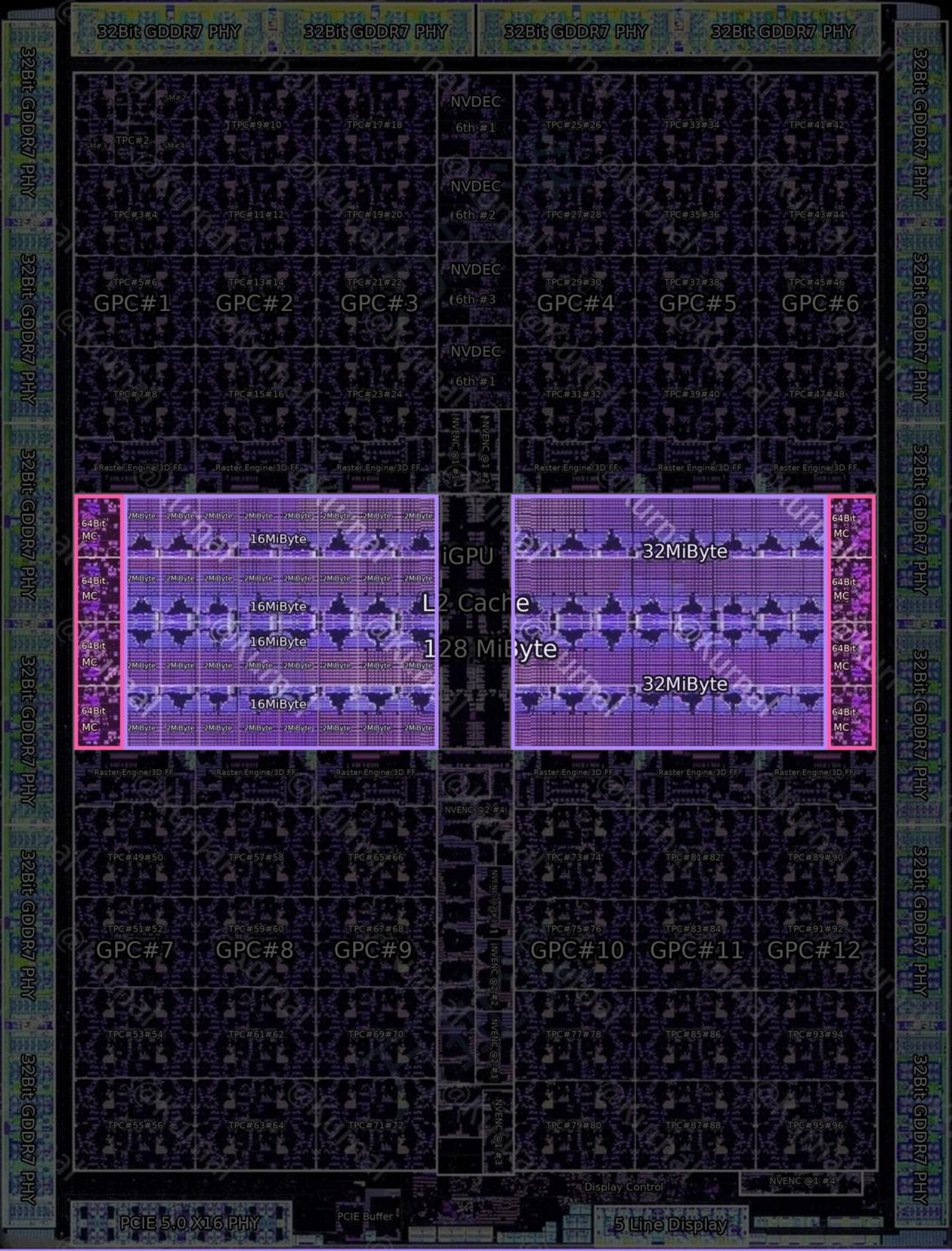
Data many SMs need again soon stays in the 128 MiB L2 (96 MB enabled on the 5090) instead of going back to memory. A bigger L2 means fewer slow trips to GDDR7.



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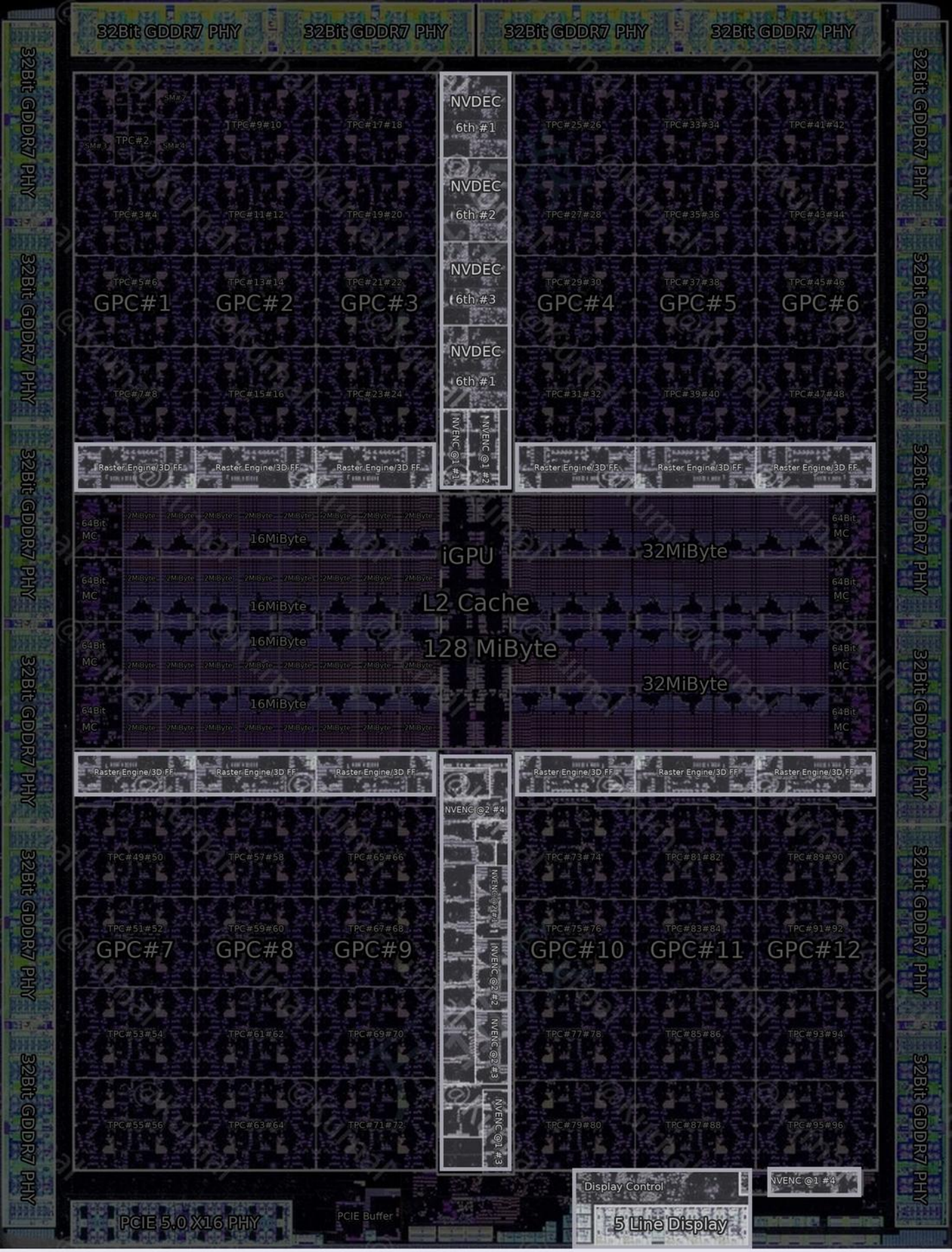
Tensor cores do the math

ML is mostly huge matrix multiplications, which tensor cores (4 per SM, 768 on the full chip) are built for. Nvidia rates the RTX 5090 at 3,352 AI TOPS at FP4. The 5090 uses 170 of the 192 SMs.



Results go back, then repeat

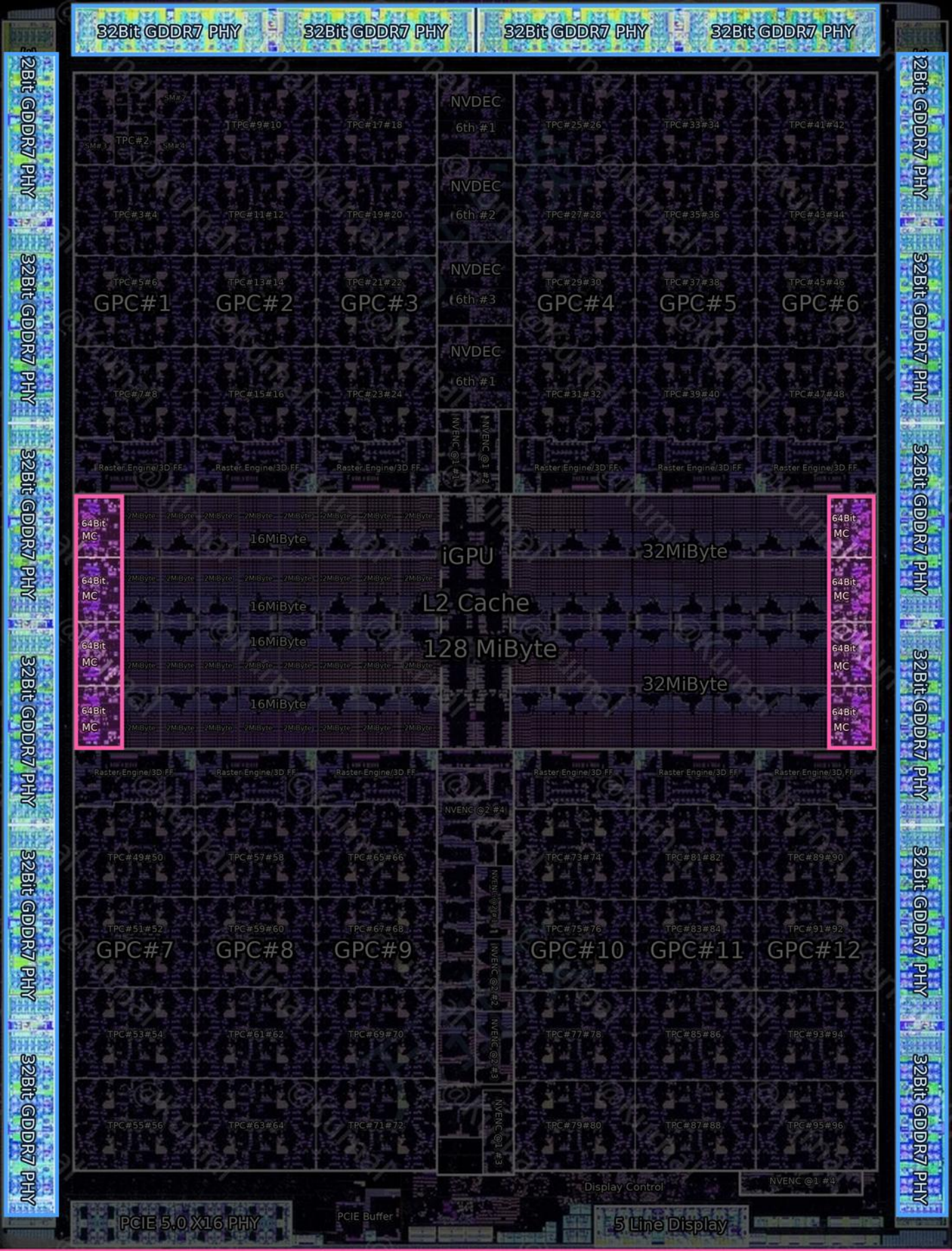
Each layer's results are written back through L2 toward memory, and the next layer starts. A language model repeats this for every layer of every token it generates.



Idle

What sits idle

Raster engines, display, and video blocks (gray) aren't used for this kind of ML. The ROPs are also idle; they live inside each GPC, and where they sit on the die isn't known.



Estimate

Why memory bandwidth sets the speed

A 7-billion-parameter model at 4 bits per weight is about 3.5 GB. If every token reads all of it once, 1.8 TB/s allows at most ~500 tokens per second. Real speeds are lower. My rough math, not a benchmark.